Adilet Shynggys

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Research Interests

Nuclear reactor thermal-hydraulics and passive safety; two-phase flow and boiling heat transfer; interfacial and multiphase fluid mechanics; capillary and contact-line dynamics; reduced-order and computational modeling of two-phase transport in confined geometries.

Education

Nazarbayev University Astana, Kazakhstan
B.Eng. in Mechanical and Aerospace Engineering *Expected June 2027*

- **GPA: 3.86 / 4.00** — ranked **1st** in cohort; Dean's List, four consecutive semesters.
- Relevant coursework: Fluid Mechanics; Engineering Thermodynamics; Numerical Methods in Engineering; Differential Equations and Linear Algebra; Applied Probability and Statistics; Engineering Dynamics I–II; Structural Mechanics; Materials and Manufacturing; CAD.
- In progress (2026–27): Heat and Mass Transfer; Mechanical Systems Design; Aerospace Propulsion.

The University of Hong Kong Hong Kong SAR
Visiting Exchange Student, Department of Mechanical Engineering *Spring 2026*

- Coursework in Energy Conversion Systems, Design and Manufacture, and Additive Manufacturing.
- Completed a multi-stage transmission design project supervised by Prof. Wei Yan — sizing shafts using the **ASME** combined bending–torsion criterion and selecting gear modules, keyways, and bearings for a specified fatigue life; separately collaborated with her on curriculum design for an undergraduate course, helping shape lecture content and structure for student engagement.
- Independent extracurricular collaboration with Prof. Homayoon Kazerooni (UC Berkeley): designed and 3D-printed flower-petal forms in SolidWorks to specified dimensional requirements, iterating through slicing and print trials to verify fit, for an interactive public art installation.

Research Experience

Undergraduate Researcher — Thermal-Hydraulics and Multiphase Flow June 2026 – Present
Department of Mechanical and Aerospace Engineering, Nazarbayev University • Advisor: Prof. Amgad Salama

- Extending a semi-analytical framework for wetting and non-wetting liquid penetration through cylindrical and tapered capillaries, resolving meniscus velocity, dynamic contact angle, and film thickness.
- Formulating and numerically integrating the coupled momentum–capillarity equations governing contact-line motion in MATLAB and Python; benchmarking against the Lucas–Washburn and Bosanquet limits.
- Developing reduced-order models of capillary-driven two-phase transport that capture the dominant interfacial physics at far lower cost than 3D simulation.
- Designing tapered SU-8 microchannels for photolithographic fabrication on silicon to test the model experimentally: the width gradient creates a Laplace-pressure asymmetry driving wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- Connecting this geometry-driven transport to passive liquid return in two-phase thermal management and to rewetting in heated channels.

Undergraduate Researcher — Triboelectric Energy Conversion Jan. – Dec. 2025
Energy Conversion Laboratory, Nazarbayev University

- Fabricated triboelectric nanogenerators by electrospinning and silicone casting; varied surface morphology and material pairing to raise measured output power density by **15–20%**.
- Built the electrical characterization workflow (open-circuit voltage, short-circuit current, load sweeps) and processed the resulting datasets in OriginPro and MATLAB.
- Contributed experimental data and analysis to a co-authored article in *Small* and to an award-winning conference poster.

Publications and Conference Presentations

[1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al. "Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator," *Small*, 2026, art. e74486. `https://doi.org/10.1002/smll.74486`

[2] **A. Shynggys** et al. "Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors," Belt and Road Initiative Conference, Astana, 2025. **Best Research Poster Award**

Engineering Design Projects

Nuclear Power Plant Conceptual Design 2026 – 2027
Senior Capstone Project, Nazarbayev University • In progress

- Developing a complete plant design from first principles: thermal power rating and heat balance, primary and secondary heat removal, and safety system architecture.
- Selecting structural, fuel, and coolant-boundary materials against thermal, mechanical, and irradiation requirements.

Ergonomic Train-Driver Seat Design 2025
*Mechaton Engineering Case Competition, sponsored by Wabtec Corporation — **1st Place***

- Designed a high-durability driver seat certified against **GOST 33330-2015**, verifying the structure under 12-hour static and dynamic fatigue loading.

Industry Experience

Product Support Engineering Intern June – July 2025
Wabtec Corporation — Kazakhstan

- Analyzed thermal performance of locomotive traction motors and supported root-cause review of field temperature excursions.
- Revised technical drawings and process documentation, cutting material waste by **10%**.

Awards and Honors

- **Presidential Scholarship of the Republic of Kazakhstan** — national merit award administered by the Ministry of Science and Higher Education 2026–2027
- **Nazarbayev University President's Scholarship** — full-tuition university merit award 2023–2027
- **Yessenov Foundation Scholarship** — selected in the **top 20** of 500+ applicants 2026–2027
- **Best Research Poster**, Belt and Road Initiative Conference 2025
- **1st Place**, Mechaton Engineering Case Competition (sponsored by Wabtec) 2025
- **Champion**, NU AIChE Engineering Challenges 2025

Technical Skills

Programming	MATLAB, Python, Mathematica, LaTeX
Modeling & Numerics	Reduced-order modeling of multiphase transport; numerical integration of coupled nonlinear ODEs; validation against analytical limiting cases
Simulation & CAD	SolidWorks (CSWA certified), OriginPro
Experimental	Electrospinning, silicone casting, 3D printing, electrical device characterization
Design Codes	ASME shaft design, GOST 33330-2015
Languages	Kazakh (native), Russian (fluent), English (IELTS Overall Band 8.0)

Teaching, Leadership, and Service

- **Physics Tutor**, Nazarbayev University — tutored 20+ first-year students. 2023–2024
- **Cohort Representative**, Mechanical and Aerospace Engineering — student–faculty liaison. 2025–2026
- **Organizer**, Charity Trail Run — 150 runners, **600,000 KZT** (∼$1,300) raised. 2025
- **Volunteer**, HKU Muslim Society and Mereke Charity Foundation. 2025–2026